

Customer Churn Analytics Using Amazon Redshift

1. Project Title

Customer Churn Analytics Using Amazon Redshift and Amazon S3

2. Objective

The objective of this project is to build a cloud-based analytics solution using Amazon Redshift to analyze customer churn data. The project involves loading telecom customer data from Amazon S3 into Amazon Redshift, transforming the data, creating analytical tables, and generating business insights related to customer churn.

3. Technologies Used

- Amazon Redshift
- Amazon S3
- AWS IAM
- Python
- Psycopg2
- Boto3
- SQL

4. AWS Resources Used

Amazon S3

Amazon S3 is used to store the raw customer churn dataset.

Dataset Location

s3://pro-123-aa/raw/sample.csv

Purpose

- Store source data
- Scalable cloud storage
- Source for Redshift COPY operation

Amazon Redshift

Amazon Redshift is used as the data warehouse for storing and analyzing customer churn data.

Cluster Configuration

Parameter	Value
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Cluster Type	Single Node
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Node Type	RA3/DC2 (depending on availability)
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Database	telecom_dw
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Region	ap-south-1
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IAM Role

IAM Role allows Redshift to access S3 securely.

Attached Policy

AmazonS3ReadOnlyAccess

Example Role

AmazonRedshift-CommandsAccessRole

6. Dataset Description

The telecom churn dataset contains customer information such as:

- Customer ID
- City
- Zip Code
- Monthly Charges
- Total Charges
- Customer Status
- Contract Information
- Internet Services
- Customer Demographics

The dataset is used to identify customer churn patterns and business trends.

7. Database Design

Table 1: churn

Stores selected customer information.

```
CREATE TABLE churn (  
    customer_id VARCHAR(50),  
    city VARCHAR(100),  
    zip_code VARCHAR(20),  
    population INT,  
    tenure BIGINT,  
    monthly_charges DECIMAL(10,2),  
    total_charges DECIMAL(10,2),  
    customer_status VARCHAR(20)  
);
```

Purpose

Stores transformed customer churn data.

Table 2: zip

Stores zip code population information.

```
CREATE TABLE zip (  
    zip_code INT,  
    population BIGINT  
);
```

Purpose

Stores population information for geographical analysis.

Table 3: churn_stage

Temporary staging table used to load raw data.

Purpose

- Initial data ingestion
- Data validation

- Data transformation

Table 4: analytics

Final analytical table used for reporting.

```
CREATE TABLE analytics (  
    customer_id VARCHAR(50),  
    city VARCHAR(100),  
    zip_code VARCHAR(20),  
    population INT,  
    tenure BIGINT,  
    monthly_charges DECIMAL(10,2),  
    total_charges DECIMAL(10,2),  
    customer_status VARCHAR(20)  
)  
  
DISTSTYLE KEY  
  
DISTKEY(zip_code)  
  
SORTKEY(tenure);
```

8. Data Loading Process

Step 1: Upload Dataset to S3

Upload the telecom churn CSV file to Amazon S3.

Example:

```
s3://pro-123-aa/raw/sample.csv
```

Step 2: Load Data into Redshift

Redshift COPY command is used for fast data loading.

```
COPY churn_stage
```

```
FROM 's3://pro-123-aa/raw/sample.csv'
```

```
IAM_ROLE 'arn:aws:iam::<ACCOUNT_ID>:role/AmazonRedshift-CommandsAccessRole'
```

```
CSV
```

```
IGNOREHEADER 1
```

```
DELIMITER ','
```

```
REGION 'ap-south-1';
```

Best Practices Followed

- CSV format used
 - Header ignored
 - Proper delimiter specified
 - IAM Role authentication used
-

Step 3: Data Transformation

Data is moved from the staging table into the final churn table.

```
INSERT INTO churn
```

```
SELECT
```

```
customer_id,
```

```
city,
```

```
zip_code,
```

```
NULL,
```

```
tenure_in_months,
```

```
monthly_charge,
```

```
total_charges,
```

```
customer_status
```

```
FROM churn_stage;
```

Benefits:

- Faster scans
- Better query execution

12. Redshift Maintenance Operations

ANALYZE

ANALYZE analytics;

Purpose

Updates table statistics used by the query optimizer.

Benefits

- Better execution plans
- Faster queries
- Improved performance

VACUUM

VACUUM analytics;

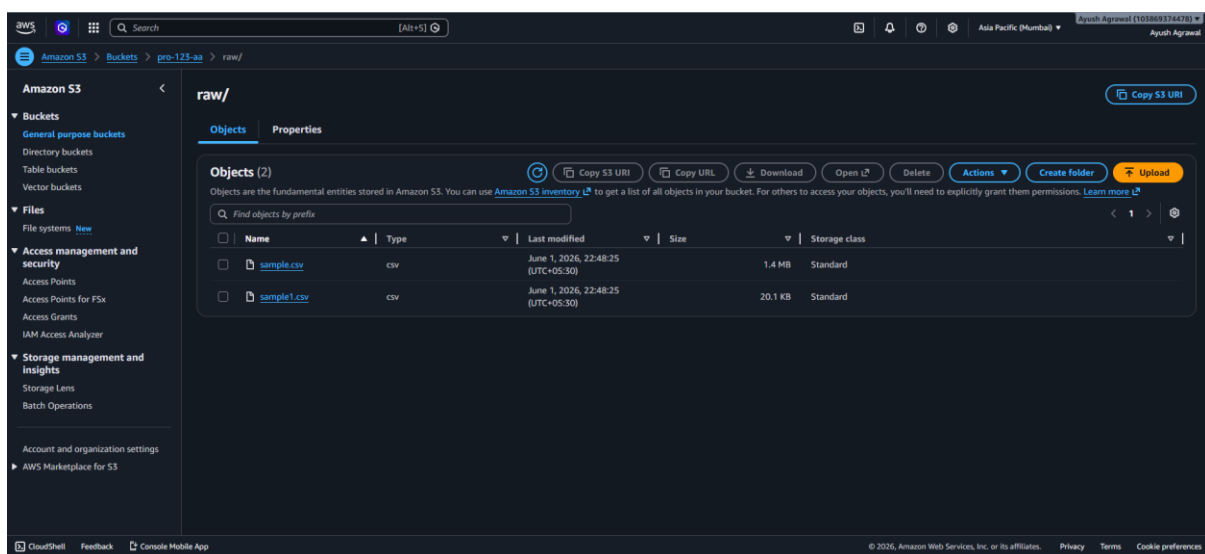
Purpose

Removes deleted records and reorganizes table storage.

Benefits

- Reclaims storage space
- Improves scan performance
- Maintains sort order

AWS S3Bucket



AWS Redshift

The screenshot displays the AWS Redshift console for a cluster named 'redshift-cluster-1'. The interface includes a top navigation bar with the AWS logo, a search bar, and user information. A blue banner at the top provides a tip about simplifying Redshift permissions management. The main content area is titled 'redshift-cluster-1' and features a 'General information' tab. This tab contains a table with the following details:

Property	Value
Cluster Identifier	redshift-cluster-1
Status	Available
Node type	ra3.large
Endpoint	redshift-cluster-1.coytydg5wwto.ap-south-1.redshift.amazonaws.com:5439/dev
Custom domain name	-
Date created	June 05, 2026, 11:02 IST
Number of nodes	1
Cluster namespace ARN	arn:aws:redshift:ap-south-1:103869374478:namespace:322cbf95-a706-4f8e-98ed-46cb1639f46c
Multi-AZ	No
Patch version	Patch 202
ODBC URL	Driver={Amazon Redshift (x64)}; Server=redshift-cluster-1.coytydg5wwto.ap-south-1.redshift.amazonaws.com; Database=dev
AWS Glue Data Catalog registration status	Registered
AWS Glue Data Catalog registration type	Amazon Redshift federated permissions
Cluster configuration	Production
Storage used	0.13 of 976.6 GB used (0.01%)
Autonomics configuration - new	Not enabled with extra compute

Below the table, there are tabs for 'Cluster details', 'Query monitoring', 'Databases', 'Datashares', 'Integrations', 'Resource Policy', 'Schedules', 'Maintenance', 'Properties', and 'Autonomics - new'. A 'Recommendations' section is also visible, indicating no recommendations are currently present. At the bottom, there are sections for 'Alarms' and 'Events'.

DataBase

This screenshot shows the 'Database connection' tab within the AWS Redshift console for the same cluster. The 'Databases' tab is selected, showing a table with the following information:

Property	Value
Cluster namespace ARN	arn:aws:redshift:ap-south-1:103869374478:namespace:322cbf95-a706-4f8e-98ed-46cb1639f46c
Multi-AZ	No
Patch version	Patch 202
ODBC URL	Driver={Amazon Redshift (x64)}; Server=redshift-cluster-1.coytydg5wwto.ap-south-1.redshift.amazonaws.com; Database=dev
AWS Glue Data Catalog registration status	Registered
AWS Glue Data Catalog registration type	Amazon Redshift federated permissions
Cluster configuration	Production
Storage used	0.13 of 976.6 GB used (0.01%)
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The 'Database connection' section shows the status as 'Connected' and provides details for the 'dev' database, including the user name 'awsuser'. A 'Change connection' button is available. Below this, there is a 'Databases (1)' section with a 'Create datashare' button.

Schemas

Database objects [info](#)
These are read-only database objects in your databases.

Schema **Tables and views** User-defined functions

Tables and views (3)

Find tables and views

	Name	Type	Database	Schema
☐	analytics	TABLE	dev	public
☐	churn	TABLE	dev	public
☐	zip	TABLE	dev	public

IAM roles

AmazonRedshift-CommandsAccessRole-20260605T110023 [info](#)

Summary

Creation date: June 05, 2026, 11:00 (UTC+05:30)

Last activity: 2 hours ago

ARN: `arn:aws:iam::103869374478:role/service-role/AmazonRedshift-CommandsAccessRole-20260605T110023`

Maximum session duration: 1 hour

Permissions Trust relationships Tags Last Accessed Revoke sessions

Permissions policies (2) [info](#)

You can attach up to 10 managed policies.

Search Filter by Type All types

☐	Policy name	Type	Attached entities
☐	AmazonRedshift-CommandsAccessPolicy-20260605T110023	Customer managed	1
☐	AmazonRedshiftAllCommandsFullAccess	AWS managed	3

Permissions boundary (not set)

Generate policy based on CloudTrail events

Project Workflow

